be located by saying how far east and how far north of Nagpur it lies. Measured on a map, Kolkata is about 960 km east and 160 km north of Nagpur, Delhi is about 190 km west and 830 km north, and Mumbai is about 650 km west and 230 km south; west and south simply count as negative amounts of east and north. Mathematicians would say that the directions east and north form a basis for the map, because they satisfy two requirements. First, every location can be reached by some combination of them; we say that they span the plane. East and west alone would not do, since no combination of them will ever take you to Delhi. Second, every location is reached by exactly one such combination; we say that they are linearly independent. Adding north-east as a third direction would spoil this, because Mumbai could then be described in endlessly many ways.

East and north are not the only directions that can do this job, either. Any two directions will serve, as long as they do not lie along the same line, as east and west do. The most convenient pairs are those at right angles to each other, like east and north, and that property will matter shortly. Turn the familiar pair through 45 degrees, for example, and we get north-east and north-west. Measured along these two new directions, Delhi is reached by going about 450 km north-east and then about 720 km north-west from Nagpur. Delhi has not moved; only the language describing its position has changed. Whichever basis we choose, it always contains exactly two directions. This number, the same for every basis of a space, is called its dimension, so our map is two-dimensional.

The state of a single qubit lives in a two-dimensional space whose coordinates are complex numbers. The vectors $|0\rangle = [1\ 0]^T$ and $|1\rangle = [0\ 1]^T$ form a basis for it, called the computational basis (also known as the standard basis or the Z basis). Writing the general qubit state as $\alpha|0\rangle + \beta|1\rangle$, as we did in the first article, simply expresses it as a combination of these two basis vectors. The computational basis is also orthonormal, and to see what that means we need the inner product. In Dirac notation, $\langle U|$ denotes the conjugate transpose of $|u\rangle$: a row vector whose entries are the complex conjugates of the entries of the column vector $|u\rangle$. The inner product $\langle u|v\rangle$ is the product of this row vector and the column vector $|v\rangle$—a single number. Two vectors are orthogonal if their inner product is zero, a vector is normalised if its inner product with itself is one, and an orthonormal basis consists of mutually orthogonal, normalised vectors. It is easy to verify that $\langle 0|1\rangle = \langle 1|0\rangle = 0$ and $\langle 0|0\rangle = \langle 1|1\rangle = 1$.

With the inner product in hand, the measurement rule from the first article can be stated in its full generality. If a qubit in the state $|\psi\rangle$ is measured in an orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$, the outcome i is obtained with probability $|\langle b_i|\psi\rangle|^2$. For the computational basis, $\langle 0|\psi\rangle = \alpha$ and $\langle 1|\psi\rangle = \beta$, and we recover the familiar probabilities $|\alpha|^2$ and $|\beta|^2$. But the rule works for any orthonormal basis. In the fourth article we met the states $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, and called them X-basis states without explaining the word. A one-line calculation shows that $\langle +|-\rangle = (1\times1 + 1\times(-1))/2 = 0$, and similar calculations give $\langle +|+\rangle = \langle -|-\rangle = 1$, so $|+\rangle$ and $|-\rangle$ form an orthonormal basis too, known as the Hadamard basis (also known as the diagonal basis or the X basis).

The letters Z and X come from the Pauli gates, which we met in the second and third articles. The Pauli-Z gate leaves $|0\rangle$ and $|1\rangle$ unchanged apart from a sign ($Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$), and the Pauli-X gate does the same for $|+\rangle$ and $|-\rangle$ ($X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$). The letters also match the Bloch sphere: the computational basis states sit at the two ends of the z-axis, the north and south poles, while the Hadamard basis states sit at the two ends of the x-axis. Note a point that confuses many beginners: orthogonal states are not at right angles on the Bloch sphere but at opposite ends of it. Figure 1 summarises both bases.

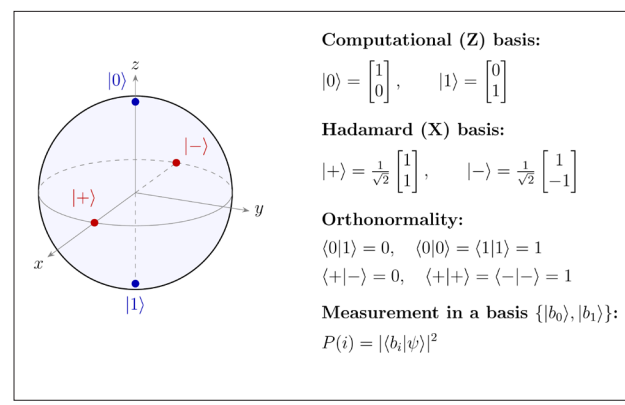


Figure 1: The computational basis and the Hadamard basis for a single qubit

Any state can be described in either basis. For example, $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ is the same state written in the language of the Hadamard basis. Rewriting a state in another basis changes only its description; applying a gate changes the state itself. All of this extends naturally to more qubits. Two qubits live in a four-dimensional space, and the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is the obvious basis for it. But in the previous article we met a second orthonormal basis for exactly the same space: the Bell basis $\{|\Phi+\rangle, |\Phi-\rangle, |\Psi+\rangle, |\Psi-\rangle\}$. Every vector in the computational basis is a product state, while every vector in the Bell basis is maximally entangled. In general, n qubits live in a space of dimension 2^n, which is why fifty qubits already need 2^{50} amplitudes—more than a quadrillion (10^{15}).

A fact from the hardware level ties all this together. The microwave readout of a superconducting qubit, described in the previous article, can answer exactly one question: is the qubit in $|0\rangle$ or in $|1\rangle$? In other words, the hardware measures only in the computational basis. To measure in any